

Cloud Application Development

Group Presentation Roadmap

Core Presentation Flow

Develop → Build → Test → Containerize → Deploy to Cloud → Monitor → Improve

The presentation should treat DevOps and Cloud Application Deployment as connected parts of one software delivery process, rather than as two unrelated topics.

1. Introduction to Cloud Application Development

What to cover: Define cloud application development as the process of designing, developing, testing, deploying and maintaining applications that use cloud infrastructure and services.

Introduce the two major areas: (1) DevOps — practices and collaboration used to build and deliver software efficiently; (2) Cloud Application Deployment — making an application available through a cloud environment.

Opening flow: Application Idea → Development → Testing → DevOps Practices → Containerization → Cloud Deployment → Monitoring → Continuous Improvement

PART I — DEVOPS

2. What is DevOps?

Explain Development (Dev), Operations (Ops), why the teams need to work together, problems with traditional software delivery, and how DevOps improves collaboration, automation and delivery.

Suggested visual: Traditional approach: Developers → Operations → Deployment Problems. DevOps: Developers ↔ Operations → Automation → Continuous Delivery → Reliable Deployment.

3. DevOps Tools

Category	Examples
Version Control	Git, GitHub, GitLab
CI/CD	Jenkins, GitHub Actions, GitLab CI/CD
Containers	Docker
Container Orchestration	Kubernetes
Infrastructure as Code	Terraform, Ansible
Monitoring	Prometheus, Grafana
Cloud Platforms	AWS, Microsoft Azure, Google Cloud

Recommended tool for detailed explanation: Docker. It connects directly to the later sections on Docker architecture, containerization and cloud deployment.

4. Components of DevOps

DevOps lifecycle: PLAN → CODE → BUILD → TEST → RELEASE → DEPLOY → OPERATE → MONITOR → back to PLAN.

- **Plan:** Define requirements, tasks and releases.
- **Code:** Write application code and manage it with version control.

- **Build:** Convert source code into a deployable application and install dependencies.
- **Test:** Automatically test the application and detect defects.
- **Release:** Prepare a tested version for deployment.
- **Deploy:** Move the application into the target environment.
- **Operate:** Keep the application running.
- **Monitor:** Track performance, errors, availability and resources.

5. Version Control

Define version control as a system for tracking changes to source code and files over time. Introduce Git and explain its value for collaboration, history, rollback and managing parallel work.

Suggested example: Main branch → feature/login + feature/payment → review → merge into main.

6. Git Branches

Explain a Git branch as an independent line of development. Demonstrate: Create Branch → Write Code → Commit → Push → Pull Request → Review → Merge.

7. CI/CD

Continuous Integration (CI): Developers frequently integrate code changes and automated builds/tests verify them.
Continuous Delivery/Deployment (CD): tested software is prepared or automatically deployed to a target environment.

8. Docker Architecture

Use the lecturer's question directly: explain the three main Docker architecture elements and how they interact.

Component	Purpose
Docker Client	Interface used to issue commands such as build, run, pull and push.
Docker Daemon / Host	Builds images, creates and manages containers, networks and volumes.
Docker Registry	Stores and distributes Docker images; Docker Hub is a common example.

Workflow: Developer → Docker build → Image → Docker push → Registry → Docker pull → Cloud server → Container.

9. Containerization vs Virtualization

Containerization	Virtualization
Uses containers	Uses virtual machines
Shares the host OS kernel	Each VM includes a guest OS
Lightweight	Generally heavier
Starts quickly	Generally takes longer to boot
Uses fewer resources	Uses more resources
Example: Docker	Examples: VMware, VirtualBox

Conclude the section by explaining why containers are useful for cloud applications: they make applications easier to package, move, deploy and scale consistently.

PART II — CLOUD APPLICATION DEPLOYMENT

10. What is Cloud Application Deployment?

Define cloud deployment as the process of making an application available on cloud infrastructure so that users can access it over a network or the Internet.

11. Cloud Deployment Process

Main pipeline: Developer → Source Code → Git Repository → CI/CD Pipeline → Build & Test → Docker Image → Container Registry → Cloud Platform → Application → Users.

Example: GitHub → GitHub Actions → Docker → Docker Hub → AWS/Azure/Google Cloud → Web Application.

12. Cloud Deployment Models

- **Public Cloud:** cloud resources provided by a cloud provider, such as AWS, Azure or Google Cloud.
- **Private Cloud:** cloud infrastructure dedicated to an organization.
- **Hybrid Cloud:** combination of private and public cloud environments.

13. Cloud Service Models

Model	What it provides	Typical focus
IaaS	Infrastructure such as virtual machines, storage and network	More infrastructure management by the customer
PaaS	Managed application platform/runtime	Deploying applications without managing all underlying infrastructure
SaaS	Complete software delivered as a service	Users consume the application

14. CI/CD in Cloud Deployment

Show how DevOps automation connects to cloud deployment: Developer → Git Push → Automated Build → Automated Tests → Build Application/Docker Image → Deploy → Cloud.

15. Practical Cloud Deployment Example

Example application: Student Management System.

- Developer writes the application.
- Code is pushed to GitHub.
- The CI/CD pipeline starts.
- The application is tested.
- A Docker image is built.
- The image is pushed to a container registry.
- A cloud server pulls the image.
- The container starts.
- Users access the application.

16. Monitoring and Maintenance

Explain that deployment is not the end of the process. Monitor CPU, memory, network, application errors, response time, availability and database performance. Mention tools such as Prometheus, Grafana, CloudWatch and Azure Monitor.

17. Security in Cloud Deployment

Briefly cover authentication, authorization, HTTPS/TLS, secrets management, firewall/access rules, regular updates and backups.

18. Advantages and Challenges

Advantages	Challenges
Scalability	Security risks
Faster deployment	Cloud costs
Automation	Configuration complexity
Reduced manual work	Internet/connectivity dependency
Collaboration	Monitoring requirements
Resource flexibility	Vendor lock-in

FINAL INTEGRATION

The final part should connect all concepts into one end-to-end story:

Application → Git → CI/CD → Build & Test → Docker → Container Registry → Cloud Deployment → Monitoring → Users

Key message: **DevOps provides the practices and automation for building and delivering software, while cloud deployment provides the infrastructure and environment where the application runs.**

Recommended Group Division

Member	Section	Main Topics
Member 1	Introduction	Cloud Application Development + DevOps introduction
Member 2	DevOps	DevOps components + lifecycle
Member 3	DevOps Tools	Git, Docker, CI/CD + Git branches
Member 4	Docker	Docker architecture + containerization vs virtualization
Member 5	Cloud Deployment	Cloud deployment, IaaS/PaaS/SaaS, CI/CD, monitoring

Recommended PowerPoint Order

1. Title
2. Learning Objectives
3. What is Cloud Application Development?
4. DevOps + Cloud Deployment Overview
5. What is DevOps?
6. Why DevOps?
7. DevOps Lifecycle
8. Components of DevOps
9. DevOps Tools
10. Version Control
11. Git Branches
12. CI/CD
13. What is Docker?
14. Docker Architecture
15. Docker Workflow
16. Containerization
17. Containerization vs Virtualization
18. Why Containers Matter in Cloud Computing
19. What is Cloud Application Deployment?
20. Cloud Deployment Process
21. Cloud Deployment Models
22. IaaS, PaaS and SaaS

- 23. Containerized Application Deployment
- 24. CI/CD Cloud Deployment
- 25. Cloud Monitoring
- 26. Cloud Security
- 27. Scalability
- 28. Advantages and Challenges
- 29. Practical Student Management System Example
- 30. Complete DevOps → Cloud Pipeline
- 31. Key Takeaways
- 32. Questions & Answers

Presentation principle: Do not present the topics as isolated definitions. Tell the story of an application from development to deployment and operation.